

Skin Intelligence & Personalized Skincare

Project Title: AI Skin Intelligence & Personalized Skincare Planner

Platform Name: Derma AI

Author: Shri Lakshmi Gurjar

Date: September 13, 2026

Version: 1.0

Executive Summary:

The AI Skin Intelligence & Personalized Skincare Planner is an AI-powered platform engineered to create highly personalized skincare routines by analyzing a user's skin profile, sleep patterns, lifestyle habits, environmental exposure, and distinct skin concerns. The platform aims to help users improve their overall skin health through advanced AI-driven skin assessments, ingredient analysis, and structured progress tracking.

By unifying data from product databases, dermatology knowledge bases, and weather/UV indices, the system provides a comprehensive, end-to-end workflow. The final deployment successfully integrates secure authentication, skin health scoring, and role-based dashboards, resulting in a production-ready application hosted via Docker on cloud infrastructure.

Introduction & Objectives

Background and Problem Statement:

The modern skincare journey lacks personalization that accounts for dynamic lifestyle and environmental factors. There is a critical need for an intelligent system that evaluates holistic skin health—incorporating elements like hydration, sleep, and UV exposure—to generate safe, effective, and adaptive daily regimens.

Target Users:

- **Consumers/Patients:** Seeking personalized routines, ingredient safety checks, and progress tracking.
- **Dermatologists:** Requiring clinical telemetry, EMR sync capabilities, and secure prescription authorization.
- **Skincare Consultants:** Needing tools to manage client profiles, monitor adherence, and adjust regimens.
- **System Administrators:** Overseeing platform analytics, user approvals, and safety audits.
- **Wellness Platforms and Beauty Brands**

Project Objectives:

- Develop an AI-powered skincare intelligence platform.
- Implement secure authentication and role-based access control.
- Build skin profile assessment and skincare recommendation workflows.
- Develop personalized skincare routine generation systems.
- Implement ingredient intelligence and product recommendation engines.
- Build skin health scoring and progress tracking modules.
- Develop dashboards for skin analytics and skincare insights.
- Deploy the platform using Docker and cloud deployment platforms such as AWS or Azure.

System Architecture

High-Level Architecture Diagram Description:

The system architecture utilizes a robust microservices approach communicating via synchronous (HTTP/HTTPS/REST) and asynchronous (Message Queue) flows. Access channels include a Responsive Web App, Mobile App, and Dashboards, which connect to an API Gateway (FastAPI) handling routing, authentication, and load balancing. The Microservices Layer processes requests and interacts with the AI/ML Intelligence Engine, while the Data Layer persists information across specialized databases.

Major Components:

- **API Gateway:** Built with FastAPI, this handles request validation, rate limiting, CORS, and authentication.
- **Microservices Layer:** Comprises independent services for Users, Skin Profiles, Assessments, Routine Planning, Ingredient Intelligence, Product Recommendations, Health Scoring, Progress Tracking, Notifications, Analytics, Reports, and Admin tasks.
- **AI/ML Engine:** A hybrid intelligence layer. It uses a custom PyTorch Multi-Modal model to extract features from facial images and fuse them with tabular lifestyle data. Google's Gemini LLM handles complex clinical reasoning and routine generation via Automatic Function Calling.
- **Data Layer:** The architecture was optimized for small-cohort, low-latency deployments. A single PostgreSQL database handles both strict relational mapping and flexible JSONB document storage, streamlining the deployment. The backend utilizes direct psycopg2 connections, which perfectly supports the target load of up to 100 concurrent users without the overhead of complex connection pooling.

Modules Overview

- 1. User Authentication & Access:** Manages JWT-based login, role assignment, and secure access control across four distinct user tiers.
- 2. Skin Profile Management:** Captures and stores skin type, age, sleep quality, hydration, allergies, and environmental exposure.
- 3. Skin Assessment Engine:** Fuses image tensors and tabular data to identify concerns, evaluate health, and calculate severity scores.
- 4. Personalized Routine Generator:** Assembles dynamic AM/PM routines and weekly treatments using active clinical synthesis.
- 5. Ingredient Intelligence Module:** Conducts interaction analysis, allergy detection, and provides educational knowledge bases for actives.
- 6. Product Recommendation Engine:** Matches commercial products to users using a 100-point rubric, optimizing for budget and ethical preferences.
- 7. Skin Health Scoring Engine:** Calculates a weighted 100-point index based on Condition, Lifestyle, Consistency, Sleep, and Hydration.
- 8. Progress Tracking:** Monitors bi-axial routine adherence vs. skin improvement over 30/60/90-day time-series windows.
- 9. Dashboards & Analytics:** Delivers highly specialized, responsive web interfaces for patients, admins, consultants, and dermatologists.
- 10. Notification System:** Triggers routine reminders, daily checklists, and alerts for adherence drop-offs.
- 11. Reports & Export System:** Features an Advanced Export Hub for generating clinical PDFs and CSV datasets for longitudinal research.
- 12. Final Integration:** Encompasses end-to-end testing, API validation, Docker containerization, and production cloud deployment.

Implementation & Tech Stack

Programming Languages:

- **Backend:** Python 3.10.
- **Frontend:** JavaScript, HTML5.

Libraries & Frameworks:

- **Backend Framework:** FastAPI, Uvicorn.
- **Frontend Frameworks:** Tailwind CSS for responsive UI design, Lucide for iconography.
- **AI/Machine Learning:** PyTorch (torch, torchvision), Scikit-learn, Pandas, NumPy. Google GenAI SDK (Gemini 3.6 Flash / 3.5 Flash-Lite).
- **Data Visualization & Tools:** Chart.js for interactive analytics. html2pdf.js for clinical report generation. DuckDuckGo Serper integration for product imagery.

Databases:

- **Primary Database:** PostgreSQL. Utilized for both relational security tables and JSONB unstructured medical payloads, eliminating the need for a secondary document store.

Deployment & DevOps:

- **Containerization:** Docker and Docker Compose.
- **Cloud Providers:** AWS / Azure.
- **Version Control:** Git, GitHub, and GitHub Actions for CI/CD.

Monitoring & Security:

- Infrastructure includes application monitoring and error tracking. Security protocols enforce strict JWT access control, parameterized SQL queries, and HIPAA/GDPR-aligned granular data privacy masking.

Testing & Results

Testing Types Executed:

- Frontend and backend integration testing.
- API validation and testing.
- End-to-end workflow testing.
- Security testing and performance optimization.

Key Performance Metrics:

- **Accuracy:** Evaluated skin concern classification accuracy, product suitability accuracy, and recommendation precision.
- **Latency & Performance:** Measured API response time, dashboard loading speed, recommendation generation latency, and concurrent user handling capacity.
- **User Satisfaction:** Tracked routine adherence accuracy, user engagement rate, and user retention rate.

Conclusion & References

Achievements:

- Successfully demonstrated a complete end-to-end skincare intelligence workflow.
- Achieved a fully deployed frontend and backend with operational dashboards and reporting systems.
- Operationalized the skincare intelligence systems, effectively generating customized skincare routines based on user profiles and goals.

Limitations & Future Scope:

- **Scalability Challenges:** Must ensure the platform can support large-scale skincare assessments and recommendation workflows while maintaining stable performance and responsiveness.
- **Future Integrations:** Plans to expand external service integration, refining NLP understanding, and expanding the localized dermatological datasets to enhance recommendation effectiveness globally.

References & Data Sources:

- APIs: Google Gemini API, DuckDuckGo Search API.
- Libraries: FastAPI, PyTorch, Chart.js, Tailwind CSS.
- Knowledge Bases: Skincare Product Databases, Ingredient Databases, Dermatology Knowledge Bases